Pair of Linear Equations in Two Variables — MCQ Worksheet

Mathematics • Chapter 3 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. A linear equation in two variables has the form:

(A) $ax + b = 0$ (B) $ax^2 + bx + c = 0$
(C) $ax + by + c = 0$ (D) $ax + by + cz = 0$

Q2. For lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to be consistent and have unique solution:

(A) $a_1/a_2 = b_1/b_2 = c_1/c_2$ (B) $a_1/a_2 \neq b_1/b_2$
(C) $a_1/a_2 = b_1/b_2 \neq c_1/c_2$ (D) Both equations are same

Q3. If $a_1/a_2 = b_1/b_2 = c_1/c_2$, the lines are:

(A) Parallel (B) Coincident (infinitely many solutions)
(C) Intersecting (D) Perpendicular

Q4. If $a_1/a_2 = b_1/b_2 \neq c_1/c_2$, the lines are:

(A) Coincident (B) Intersecting
(C) Parallel (no solution) (D) Perpendicular

Q5. Geometrically, $x - y = 2$ and $x + y = 4$ are:

(A) Parallel (B) Intersecting at (3, 1)
(C) Coincident (D) Perpendicular only

Q6. Solution of $x + y = 7$, $x - y = 3$ is:

(A) (2, 5) (B) (5, 2)
(C) (3, 4) (D) (4, 3)

Q7. Solution of $2x + y = 8$, $x - y = 1$ is:

(A) (3, 2) (B) (2, 3)
(C) (1, 6) (D) (0, 8)

Q8. The pair $x + 2y = 5$, $3x + 6y = 15$ has:

(A) No solution (B) Unique solution
(C) Infinitely many solutions (D) Two solutions

Q9. The pair $x + y = 3$, $2x + 2y = 7$ has:

(A) Unique solution (B) Infinitely many solutions
(C) No solution (D) Exactly two

Q10. For $3x - ky = 6$ and $6x - 4y = 12$ to have infinitely many solutions, k =

(A) 1 (B) 2
(C) 3 (D) 4

Q11. The substitution method involves:

(A) Substituting one variable in terms of another (B) Multiplying both equations
(C) Comparing slopes (D) Cross-multiplication

Q12. Elimination method works by:

(A) Removing one variable by adding/subtracting (B) Substituting
(C) Drawing graph (D) Using cross multiplication formula

Q13. Cross-multiplication method gives x and y in terms of:

(A) a_1, b_1, c_1 etc (B) Sum of coefficients
(C) Slopes (D) Roots

Q14. The sum of two numbers is 25 and difference is 7. The numbers are:

(A) 16, 9 (B) 18, 7
(C) 20, 5 (D) 15, 10

Q15. The cost of 5 pencils and 3 pens is Rs 35; cost of 3 pencils and 5 pens is Rs 37. Cost of 1 pencil and 1 pen is:

(A) Rs 9
(B) Rs 8
(C) Rs 11
(D) Rs 12

Q16. If 5 boys and 4 girls cost Rs 100 of fruit and 4 boys and 5 girls cost Rs 110, then per boy & per girl is:

(A) Rs 8 and Rs 15
(B) Rs 5 and Rs 10
(C) Rs 10 and Rs 12.5
(D) Rs 5 and Rs 18.75

Q17. For $2x + 3y = 7$ and $(a – b)x + (a + b)y = 3a + b – 2$ to have infinitely many solutions:

(A) $a = 5, b = 1$
(B) $a = 1, b = 5$
(C) $a = 2, b = 3$
(D) $a = 0, b = 0$

Q18. Two lines are coincident if they have:

(A) Different slopes
(B) Same slope but different intercept
(C) Same slope and same intercept
(D) Slopes adding to 0

Q19. If we plot $2x + 3y = 12$ and $4x + 6y = 24$, lines are:

(A) Parallel
(B) Coincident
(C) Intersecting
(D) Perpendicular

Q20. Half the perimeter of a rectangular garden is 36 m. If length is 4 m more than width, the dimensions are:

(A) 20 m × 16 m
(B) 18 m × 14 m
(C) 22 m × 14 m
(D) 16 m × 12 m

Q21. In a fraction, if numerator is reduced by 1 it becomes 1/3, and if denominator is increased by 8 it also becomes 1/3. The fraction is:

(A) 3/8
(B) 2/5
(C) 5/12
(D) 4/12

Q22. A boat goes 30 km upstream in 5 h and 30 km downstream in 3 h. Speed of stream is:

(A) 1 km/h
(B) 2 km/h
(C) 3 km/h
(D) 4 km/h

Q23. The age of father is 3 times that of son. After 12 years it will be twice. Son's present age is:

(A) 10
(B) 12
(C) 14
(D) 16

Q24. For unique solution we need:

(A) Determinant of coefficient matrix non-zero ($a_1b_2 – a_2b_1 \neq 0$)
(B) $a_1 = a_2$
(C) $b_1 = b_2$
(D) $c_1 = c_2$

Q25. Lines having same y-intercept can still:

(A) Have different slopes
(B) Always be coincident
(C) Always be parallel
(D) Always meet at origin

— End of Worksheet —